

APPENDIX

In this supplementary, we provide additional details and results that supplement CONTACTMIMIC:

1. § A describes the organization of the supplementary video and the anonymous project page.
2. § B provides additional implementation details, including PPO hyperparameters, policy observations and architecture, and domain randomization.
3. § C lists the full set of reward terms with their weights and shaping parameters.
4. § D details our paired-motion generation pipeline: contact-label extraction, augmentation parameters, and the per-motion training configuration.
5. § E describes our real-world deployment on the Unitree G1.
6. § F reports additional quantitative results, including the full per-motion contact controllability table and the per-motion data augmentation ablation.

A Supplementary Videos and Project Page

We provide an overview video and the raw recordings of every real-world trial reported in Table 2 in the supplementary materials, and an anonymous [project page](#) that we strongly encourage reviewers to view before reading this appendix. The videos and page include:

- **Method overview:** the full pipeline, from human-motion retargeting and paired-motion augmentation that breaks keypoint–contact correlations, to contact-conditioned policy training with our contact-following rewards and real-world deployment.
- **Real-world results:** for each of the five tested motions (*wipe whiteboard*, *sit in front of table*, *lean on backrest I/II*, *sit and squat*), side-by-side clips of the same policy on the real Unitree G1 commanded with contact ✓ and contact ✗, demonstrating on-command toggling of physical contact.
- **Simulation results:** per-motion clips across all 10 motions under the four eval-time trajectories $\mathcal{T}_{\text{near}}^{\checkmark} - \mathcal{T}_{\text{far}}^{\checkmark}$, including the flipped conditions $\mathcal{T}_{\text{near}}^{\times}$ and $\mathcal{T}_{\text{far}}^{\checkmark}$ that test pure contact-label following.
- **Baseline comparison:** side-by-side comparisons against the state-of-the-art keypoint-only tracker BeyondMimic [1] under the same keypoint trajectory, showing that without contact conditioning it can only follow the single mode baked into the trajectory and cannot be steered at test time.
- **Ablation:** comparisons against a no-augmentation variant, showing that without our paired-motion augmentation the policy ignores the contact command and fails contact-controlled tasks (e.g., *sit on table*, *lean on backrest II*).

B Implementation Details

We train one policy per motion with PPO [44] in Isaac Lab [43] across 4096 parallel environments at 50 Hz. At each environment reset, the episode is randomly assigned either the default trajectory or one of its augmented counterparts (§ 3.2), so the same policy sees both contact and no-contact target labels during training. PPO hyperparameters are listed in Table 5.

Policy Observations. The actor observation $\mathbf{o}_t$ concatenates: (i) *proprioception* $\mathbf{p}_t$: joint positions, joint velocities, base angular velocity, and projected gravity; (ii) *reference keypoint targets* $\bar{\mathbf{k}}_t$, expressed in the robot’s local frame and derived from the reference configuration $\bar{\mathbf{q}}_t$ via forward kinematics; (iii) the *binary reference contact label* $\bar{\mathbf{c}}_t \in \{0, 1\}^{|\mathcal{B}|}$ (with $\bar{c}_{t,b} = \max_p \bar{c}_{t,b,p}$) for the current frame. Uniform observation noise is added during training to all actor observations.

Policy Architecture. Both actor and critic are MLPs with hidden dimensions [512, 256, 128] and ELU activations. The critic receives *noise-free* versions of the actor’s observations, plus the base linear velocity, which is not available to the actor; this acts as the critic’s privileged information.

Table 5: PPO hyperparameters used to train each contact-conditioned policy.

Parameter	Value
Parallel environments	4096
Policy step rate	50 Hz
Steps per env per update	24
Max iterations	30,000
Learning rate	1×10^{-3} (adaptive, KL target 0.01)
Discount factor γ	0.99
GAE λ	0.95
PPO clip parameter	0.2
Mini-batches per epoch	4
Learning epochs per update	5
Entropy coefficient	5×10^{-3}
Value-loss coefficient	1.0
Max gradient norm	1.0
Initial action noise std	1.0

Domain Randomization. We randomize, at each episode start, robot link masses, joint friction, and object friction and mass within standard ranges, to facilitate sim-to-real transfer.

C Reward Details

Following § 3.1, the per-step reward decomposes into tracking, contact-aware, and regularization terms. Table 6 lists every reward term, its functional form, weight, and any tunable shaping parameter. Tracking terms follow BeyondMimic [1] and use Gaussian kernels $\exp(-e^2/\sigma^2)$ on the corresponding error signal e .

Table 6: Full list of reward terms used to train each contact-conditioned policy. Positive weights are rewards; negative weights are penalties. Gaussian terms are shaped as $\exp(-e^2/\sigma^2)$ on the underlying error e .

Group	Term	Error signal / form	Weight
Tracking	Global anchor pos.	Root position error; $\sigma = 0.3$ m	0.5
	Global anchor ori.	Root orientation error; $\sigma = 0.4$ rad	0.5
	Body position	Per-body position error; $\sigma = 0.3$ m	1.0
	Body orientation	Per-body orientation error; $\sigma = 0.4$ rad	1.0
	Body lin. vel.	Per-body linear velocity error; $\sigma = 1.0$ m/s	1.0
	Body ang. vel.	Per-body angular velocity error; $\sigma = \pi$ rad/s	1.0
Contact	Label matching	Balanced accuracy $\frac{1}{2}(\text{TPR} + \text{TNR})$, force threshold 1 N	4.0
	Distance	Gaussian on body-to-part dist., $\sigma = 0.2$ m; unwanted-contact penalty at $\delta = 0.05$ m	3.0
Reg.	Action rate	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.15
	Joint limits	Quadratic, on joint-limit violation	-10.0
	Undesired contacts	Non-foot/wrist body in contact with terrain	-0.1

The contact label-matching reward defaults to the balanced-accuracy form $r_t^{\text{lm}} = \frac{1}{2}(\text{TPR} + \text{TNR})$; for sparse-contact motions we instead use the TP-FP form $r_t^{\text{lm}} = \text{TPR} - \lambda \text{FPR}$ with $\lambda = 1.0$, which provides a stronger gradient when most $\bar{c}_{t,b,p} = 0$ and TNR saturates.

D Paired Motion Generation Details

Contact label extraction. Given a retargeted trajectory from OmniRetarget [29], we replay the trajectory in MuJoCo and at each frame t mark a robot body part b as in contact ($\bar{c}_{t,b,p} = 1$) when its distance to the object surface is below 1 cm; the semantic part p is assigned by looking up the nearest object surface point.

Augmentation parameters. We synthesize three augmentations (§ 3.2). For inflated-geometry augmentation, the collision geometry of the target object part(s) is expanded outward by an isotropic offset of $\delta_{\text{infl}} \in [5, 10]$ cm in all directions during a second retargeting pass; this forces the retargeter to route the relevant robot bodies further from the original surface, and the contact labels of the target pairs are zeroed out accordingly. At training time we further apply random contact-label flipping and random interaction-object removal; the per-motion probabilities vary slightly and are listed in Table 7.

Per-motion settings. Table 7 summarizes the per-motion training configuration: which augmentations are used, the contact label-matching reward mode (balanced accuracy or TP–FP), the augmentation probabilities, and the number of PPO iterations. Sparse-contact motions (e.g., *wipe whiteboard*, free-object kicks and lifts) use the TP–FP mode to provide a stronger gradient when most reference labels are zero.

Table 7: Per-motion training settings. *Aug.* columns indicate which paired-motion augmentations are enabled: *Infl.* (inflated geometry), *Rem.* (interaction-object removal at train time), and *Flip* (random contact-label flipping). *Mode* is the label-matching reward form (Bal. = balanced accuracy, TP–FP = TPR – λ FPR, $\lambda=1$). $p_{\text{rem.}}$ and p_{flip} are the train-time augmentation probabilities (per episode).

Motion	Infl. aug.	Rem. aug.	Flip aug.	Mode	$p_{\text{rem.}}/p_{\text{flip}}$
Wipe whiteboard	✓	✓	✓	TP–FP	0.3 / 0.2
Sit in front of table	✓	✓	✓	Bal.	0.3 / 0.2
Lean on backrest I	✓	✓	✓	Bal.	0.3 / 0.2
Lean on backrest II	✓	✓	✓	Bal.	0.3 / 0.2
Step foot on chair	✓	✓	✓	Bal.	0.3 / 0.2
Sit on table	✓	✗	✓	Bal.	0.0 / 0.2
Lean against table	✓	✓	✓	Bal.	0.3 / 0.2
Sit and squat	✗	✓	✓	Bal.	0.3 / 0.2
Kick chair†	✓	✓	✓	TP–FP	0.3 / 0.2
Pick up box†	✓	✓	✓	TP–FP	0.3 / 0.2

Sit and squat is the only motion that does *not* use inflated geometry: because the augmented “no-contact” trajectory for this motion is simply a squat at the same height, we obtain it directly by removing the chair from the scene rather than by re-retargeting against an inflated chair. *Sit on table* skips the object-removal augmentation since the robot needs to sit on the table surface, and removing the table would directly lead to failure.

E Real-World Deployment

We deploy each policy on a Unitree G1 (29 actuated DoFs). The policy runs onboard at 50 Hz, producing target joint positions that are tracked by the robot’s joint-level PD controller at 1000 Hz, identical to the simulation control stack. Object placement (chair, table, board, bench) is measured by hand before each trial and reproduced on a fixed mat to align the robot’s start pose with the reference. No external motion capture or vision is used at deployment time; the policy receives only its onboard proprioception, the pre-recorded reference, and the contact-label command.

F Additional Quantitative Results

Per-motion contact controllability. Table 8 reports the full per-motion contact controllability results across all four trajectories ($\mathcal{T}_{\text{near}}^{\checkmark} - \mathcal{T}_{\text{far}}^{\checkmark}$). Contact bodies and impulse track the contact label rather than the keypoint variant, confirming the policy follows the commanded contact.

Per-motion takeaways from Table 8. While the overall trend—contact bodies and impulse track the contact label—holds across motions, several per-motion patterns are worth highlighting:

- **Sit on table** shows the cleanest separation: when commanded contact ✗, both impulse and bodies collapse to near zero ($\mathcal{T}_{\text{far}}^{\times}, \mathcal{T}_{\text{near}}^{\times}$), while contact ✓ recovers them to the reference level ($\mathcal{T}_{\text{near}}^{\checkmark}, \mathcal{T}_{\text{far}}^{\checkmark}$). This is the prototypical case where the contact label fully drives the policy’s behavior.

- **Step foot on chair** is an interesting partial-control case: contact bodies are essentially identical across all four trajectories (~ 0.6), because the robot cannot balance on a single leg with the other foot raised that high, so it *must* keep contact with the chair. However, the contact impulse still tracks the label ($\mathcal{T}_{\text{near}}^{\mathbf{x}}$ 1.16 N·s vs. $\mathcal{T}_{\text{near}}^{\mathbf{v}}$ 1.62 N·s), showing that the policy modulates how *firmly* the foot pushes down even when contact is unavoidable.
- **Lean on backrest I** and **Wipe whiteboard** show some residual contact under contact $\mathbf{x}$ with near keypoints ($\mathcal{T}_{\text{near}}^{\mathbf{x}}$ bodies 0.68 and 0.52): the reference keypoints place the torso/hand sufficiently close to the surface that fully suppressing contact would incur a keypoint tracking error whose cost exceeds the contact-aware penalty under our reward weighting, leading the policy to settle at a trade-off where residual contact persists. The impulse, however, still drops by a factor of $3\text{--}4\times$.
- For the free-object motions, the most informative signal is the object displacement. **Kick chair** only produces a brief foot–chair contact (two kicks), so contact bodies and impulse are tiny (~ 0.01) and barely vary across conditions. Yet the displacement shows that with contact $\mathbf{v}$ and near keypoints ($\mathcal{T}_{\text{near}}^{\mathbf{v}}$) the two kicks land cleanly (0.31 m), while contact $\mathbf{x}$ ($\mathcal{T}_{\text{near}}^{\mathbf{x}}$, 0.26 m) or far keypoints with contact $\mathbf{v}$ ($\mathcal{T}_{\text{far}}^{\mathbf{v}}$, 0.17 m) lead to weaker or partial contact.
- **Pick up box** has the largest displacement under contact $\mathbf{v}$ ($\mathcal{T}_{\text{near}}^{\mathbf{v}}$ 0.49, $\mathcal{T}_{\text{far}}^{\mathbf{v}}$ 0.50): the policy commits to grasping and lifting the box. Under contact $\mathbf{x}$ the box is barely displaced ($\mathcal{T}_{\text{far}}^{\mathbf{x}}$ 0.18, $\mathcal{T}_{\text{near}}^{\mathbf{x}}$ 0.20), even though the keypoint trajectory for $\mathcal{T}_{\text{near}}^{\mathbf{x}}$ is the original pickup motion—this is the clearest example of contact conditioning enabling task-relevant manipulation that keypoints alone cannot produce.

Data augmentation ablation (per-motion). Fig. 6 visualizes the per-motion arrow plot for both contact bodies and impulse, and Table 9 reports the underlying numbers for our method with and without the data augmentation strategies from § 3.2, evaluated on $\mathcal{T}_{\text{near}}^{\mathbf{v}}$ and $\mathcal{T}_{\text{near}}^{\mathbf{x}}$. Without augmentation, contact metrics are largely insensitive to the contact label ($\mathcal{T}_{\text{near}}^{\mathbf{v}}$ and $\mathcal{T}_{\text{near}}^{\mathbf{x}}$ produce similar values), showing the policy ignores the conditioning signal.

Per-motion takeaways from Fig. 6 and Table 9. The augmentation gap is largest on motions where contact is geometrically *not forced* by the keypoint reference:

- **Sit and squat** shows the most dramatic effect: under contact $\mathbf{x}$ ($\mathcal{T}_{\text{near}}^{\mathbf{x}}$), No-aug still produces 0.46 contact bodies and 1.17 N·s impulse (nearly identical to its $\mathcal{T}_{\text{near}}^{\mathbf{v}}$ value of 0.47/1.57), meaning the policy ignores the contact label entirely; with augmentation, the same $\mathcal{T}_{\text{near}}^{\mathbf{x}}$ drops to 0.08/0.05, roughly a $6\text{--}23\times$ reduction.
- **Sit in front of table**, **Sit on table**, and **Lean on backrest II** follow the same pattern: No-aug barely modulates contact between $\mathcal{T}_{\text{near}}^{\mathbf{v}}$ and $\mathcal{T}_{\text{near}}^{\mathbf{x}}$, while Ours produces a clear ON/OFF separation.
- **Step foot on chair**, by contrast, has nearly identical numbers for No-aug and Ours, because the keypoint reference itself forces foot contact regardless of the label—there is little room for augmentation to teach contact decoupling here.
- **Pick up box** shows that without augmentation the policy keeps grabbing the box even under contact $\mathbf{x}$ (No-aug $\mathcal{T}_{\text{near}}^{\mathbf{x}}$: 1.95 bodies, 1.85 N·s); augmentation cuts these to 0.29/0.17, allowing the same policy to follow a “don’t touch” command on a manipulation motion.
- **Kick chair** is unaffected by the ablation (both variants ~ 0.01): the contact is so brief and the keypoint trajectory so determined by the kick swing that there is essentially no contact-vs-no-contact decision for the augmentation to teach.

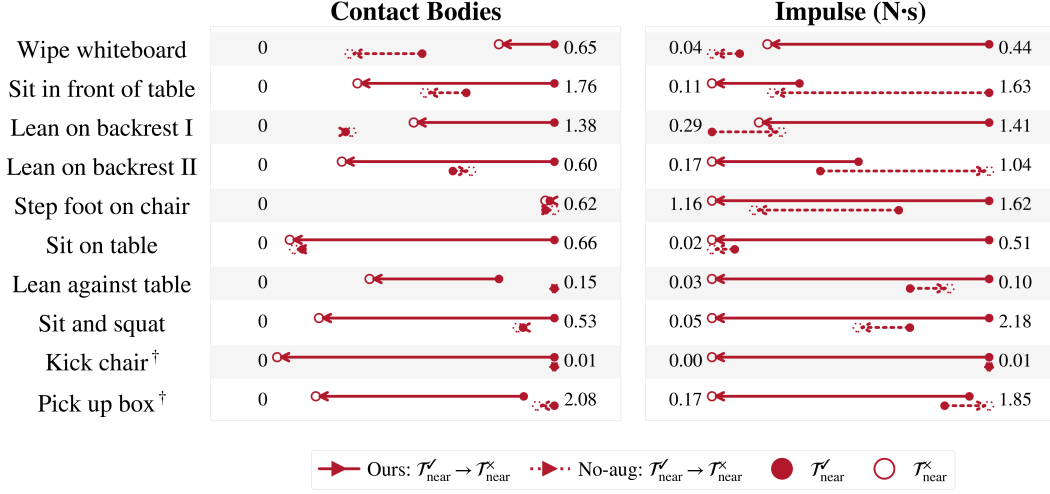


Figure 6: Per-motion ablation on paired-motion data augmentation, shown for both contact bodies and impulse. **Solid arrows** show our full method (Ours); **dotted arrows** show the no-augmentation baseline (No-aug). Both arrows go from $\mathcal{T}_{\text{near}}^{\checkmark}$ to $\mathcal{T}_{\text{near}}^{\times}$. Without augmentation, the arrows are short or even reverse direction, indicating the policy fails to follow the contact label.

Table 9: Ablation on paired-motion data augmentation across all motions. ($\mathcal{T}_{\text{near}}^{\checkmark}$ = original keypoint with contact; $\mathcal{T}_{\text{near}}^{\times}$ = original keypoint without contact (flipped)). Without augmentation, contact metrics are largely insensitive to the contact label (both $\mathcal{T}_{\text{near}}^{\checkmark}$ and $\mathcal{T}_{\text{near}}^{\times}$ produce similar values, whether high or low), showing the policy ignores the contact conditioning.

Motion	Variant	Contact bodies		Impulse (N·s)	
		$\mathcal{T}_{\text{near}}^{\checkmark}$	$\mathcal{T}_{\text{near}}^{\times}$	$\mathcal{T}_{\text{near}}^{\checkmark}$	$\mathcal{T}_{\text{near}}^{\times}$
Wipe whiteboard	No-aug	0.34±0.40	0.17±0.24	0.08±0.12	0.04±0.08
	Ours	0.65±0.45	0.52±0.44	0.44±0.37	0.12±0.13
Sit in front of table	No-aug	1.20±0.44	0.94±0.49	1.63±1.50	0.44±0.42
	Ours	1.76±0.33	0.51±0.64	0.59±0.22	0.11±0.21
Lean on backrest I	No-aug	0.34±0.51	0.37±0.40	0.29±0.66	0.57±1.19
	Ours	1.38±1.27	0.68±0.65	1.41±1.46	0.48±0.64
Lean on backrest II	No-aug	0.38±0.39	0.42±0.45	0.51±0.69	1.04±1.42
	Ours	0.60±0.39	0.14±0.24	0.63±0.55	0.17±0.37
Step foot on chair	No-aug	0.60±0.45	0.62±0.48	1.47±1.19	1.23±1.02
	Ours	0.61±0.48	0.60±0.48	1.62±1.33	1.16±1.01
Sit on table	No-aug	0.06±0.17	0.04±0.14	0.06±0.48	0.02±0.16
	Ours	0.66±0.47	0.03±0.16	0.51±0.60	0.02±0.20
Lean against table	No-aug	0.15±0.43	0.15±0.43	0.08±0.23	0.09±0.25
	Ours	0.12±0.30	0.05±0.22	0.10±0.51	0.03±0.16
Sit and squat	No-aug	0.47±0.50	0.46±0.46	1.57±1.66	1.17±1.72
	Ours	0.53±0.51	0.08±0.22	2.18±2.22	0.05±0.18
Kick chair†	No-aug	0.01±0.06	0.01±0.07	0.01±0.27	0.01±0.12
	Ours	0.01±0.08	0.00±0.05	0.01±0.12	0.00±0.06
Pick up box†	No-aug	2.08±0.92	1.95±0.93	1.58±1.13	1.85±1.26
	Ours	1.85±0.94	0.29±0.63	1.73±1.18	0.17±0.59

Table 8: Contact label controllability across all interaction motions. A single contact-conditioned policy is evaluated on four input trajectories per motion: $\mathcal{T}_{\text{near}}^\checkmark$ and $\mathcal{T}_{\text{far}}^\times$ are matched (keypoint and label agree), while $\mathcal{T}_{\text{near}}^\times$ and $\mathcal{T}_{\text{far}}^\checkmark$ are flipped (keypoint and label disagree). Object displacement is only reported for free-object motions ($\dagger$). Bodies/impulse track the contact label (not the keypoint variant), indicating the policy follows the label rather than implicitly inferring contact from keypoints. The arrow beside each key joint marks whether its torque is intuitively expected to increase ($\uparrow$) or decrease ($\downarrow$) when the commanded contact is turned off.

Motion	Trajectory	Contact bodies	Ref	Impulse (N·s)	Key torque (N·m)
Wipe whiteboard	$\mathcal{T}_{\text{near}}^\checkmark$	0.65 ± 0.45	0.34	0.44 ± 0.37	-2.07 ± 1.61 (R wrist pitch $\uparrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.25 ± 0.25	0.00	0.04 ± 0.06	-0.44 ± 0.30 (R wrist pitch $\uparrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.52 ± 0.44	0.34	0.12 ± 0.13	-0.76 ± 0.60 (R wrist pitch $\uparrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.33 ± 0.39	0.00	0.07 ± 0.11	-0.53 ± 0.46 (R wrist pitch $\uparrow$)
Sit in front of table	$\mathcal{T}_{\text{near}}^\checkmark$	1.76 ± 0.33	2.68	0.59 ± 0.22	-1.46 ± 6.26 (waist pitch $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.08 ± 0.19	0.00	0.01 ± 0.09	-12.12 ± 5.08 (waist pitch $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.51 ± 0.64	2.68	0.11 ± 0.21	-14.91 ± 7.54 (waist pitch $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.57 ± 0.46	0.00	0.58 ± 2.70	-4.65 ± 8.78 (waist pitch $\downarrow$)
Lean on backrest I	$\mathcal{T}_{\text{near}}^\checkmark$	1.38 ± 1.27	2.57	1.41 ± 1.46	19.2 ± 8.4 (hip $\uparrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.51 ± 0.49	1.36	0.16 ± 0.27	29.4 ± 17.3 (hip $\uparrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.68 ± 0.65	2.57	0.48 ± 0.64	24.1 ± 12.5 (hip $\uparrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.60 ± 0.48	1.36	0.28 ± 0.26	26.7 ± 13.3 (hip $\uparrow$)
Lean on backrest II	$\mathcal{T}_{\text{near}}^\checkmark$	0.60 ± 0.39	0.70	0.63 ± 0.55	26.7 ± 12.0 (hip $\uparrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.00 ± 0.02	0.00	0.00 ± 0.06	40.7 ± 18.9 (hip $\uparrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.14 ± 0.24	0.70	0.17 ± 0.37	32.2 ± 16.6 (hip $\uparrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.02 ± 0.11	0.00	0.03 ± 0.27	40.8 ± 21.1 (hip $\uparrow$)
Step foot on chair	$\mathcal{T}_{\text{near}}^\checkmark$	0.61 ± 0.48	0.64	1.62 ± 1.33	$+0.10 \pm 11.62$ (L hip pitch $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.62 ± 0.48	0.00	1.60 ± 1.31	-26.66 ± 19.59 (L hip pitch $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.60 ± 0.48	0.64	1.16 ± 1.01	-4.13 ± 14.82 (L hip pitch $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.61 ± 0.48	0.00	1.59 ± 1.32	-2.77 ± 11.17 (L hip pitch $\downarrow$)
Sit on table	$\mathcal{T}_{\text{near}}^\checkmark$	0.66 ± 0.47	0.53	0.51 ± 0.60	$+2.43 \pm 5.29$ (waist roll $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.04 ± 0.18	0.00	0.03 ± 0.25	-5.99 ± 4.97 (waist roll $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.03 ± 0.16	0.53	0.02 ± 0.20	-5.91 ± 5.01 (waist roll $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.67 ± 0.46	0.00	0.48 ± 0.43	$+1.61 \pm 5.29$ (waist roll $\downarrow$)
Lean against table	$\mathcal{T}_{\text{near}}^\checkmark$	0.12 ± 0.30	0.21	0.10 ± 0.51	8.4 ± 6.4 (waist pitch $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.05 ± 0.21	0.02	0.03 ± 0.16	7.8 ± 8.7 (waist pitch $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.05 ± 0.22	0.21	0.03 ± 0.16	7.9 ± 8.9 (waist pitch $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.13 ± 0.43	0.02	0.09 ± 0.32	8.0 ± 9.0 (waist pitch $\downarrow$)
Sit and squat	$\mathcal{T}_{\text{near}}^\checkmark$	0.53 ± 0.51	1.28	2.18 ± 2.22	19.0 ± 7.5 (hip $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.01 ± 0.05	1.28	0.02 ± 0.24	15.1 ± 6.9 (hip $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.08 ± 0.22	1.28	0.05 ± 0.18	11.4 ± 4.7 (hip $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.81 ± 0.80	1.28	1.88 ± 2.19	13.8 ± 5.3 (hip $\downarrow$)
Kick chair $\dagger$	$\mathcal{T}_{\text{near}}^\checkmark$	0.01 ± 0.08	0.09	0.01 ± 0.12	0.31 ± 0.15 (disp. $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.01 ± 0.08	0.04	0.01 ± 0.18	0.16 ± 0.13 (disp. $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.00 ± 0.05	0.09	0.00 ± 0.06	0.26 ± 0.10 (disp. $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	0.01 ± 0.06	0.04	0.01 ± 0.09	0.17 ± 0.10 (disp. $\downarrow$)
Pick up box $\dagger$	$\mathcal{T}_{\text{near}}^\checkmark$	1.85 ± 0.94	1.89	1.73 ± 1.18	0.49 ± 0.47 (disp. $\downarrow$)
	$\mathcal{T}_{\text{far}}^\times$	0.36 ± 0.68	0.00	0.21 ± 0.72	0.18 ± 0.26 (disp. $\downarrow$)
	$\mathcal{T}_{\text{near}}^\times$	0.29 ± 0.63	1.89	0.17 ± 0.59	0.20 ± 0.24 (disp. $\downarrow$)
	$\mathcal{T}_{\text{far}}^\checkmark$	1.22 ± 1.20	0.00	1.17 ± 1.40	0.50 ± 0.46 (disp. $\downarrow$)

$\dagger$ Free-object motion; the torque column reports object displacement (disp., m) instead.